

The effect of short videos on obesity: Evidence from China

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ABSTRACT

Obesity rates have surged in recent years, particularly in developing countries, yet key drivers remain inadequately understood. We identify short videos as a novel and significant determinant of obesity. We use China Family Panel Studies (CFPS) microdata in China to examine the causal relationship between short videos and obesity by the fixed effect panel regression model. Our findings show that short videos increase BMI by 0.12 and the probability of obesity by 2.2 %, resulting in an additional CNY of 7.52 billion one year in healthcare costs nationwide. It can explain about 27 % of the growth in China's obese population from 2020 to 2022, suggesting that it has become a major new driver of obesity. Further analysis reveals that short videos are addictive, displacing time for physical exercise, and we find no evidence supporting their positive impact on exercise by information channel. Other potential mechanisms, including mental health, sleep quality, and health-related habits, are found to be insignificant. Heterogeneity analysis indicates that men, individuals with lower education levels, those aged over 35, and urban residents are more likely to experience negative effects from short video addiction. This study contributes to the literature by highlighting short video platforms as emerging digital determinants of obesity, providing new perspectives for health policy and digital media regulation.

1. Introduction

Obesity rates have surged in recent years, particularly over the past decade. Nearly two in five adults are now classified as obese, and by 2030, half of the global adult population is projected to be obese (Ng et al., 2025). Obesity is a major risk factor for diabetes, cardiovascular disease, and cancer, and it significantly increases the risk of premature death (Sun et al., 2022). Moreover, it imposes substantial economic costs, with an estimated annual loss equivalent to 2.19 % of global GDP (Okunogbe et al., 2022). Given its extensive health and economic implications, the obesity epidemic has emerged as a critical issue on global public health agendas.

Some studies have predominantly examined factors such as food prices (Anderson and Matsa, 2011; Cawley, 2015; Christian and Rashad, 2009; Courtemanche and Carden, 2011; Currie et al., 2010), income (Akee et al., 2013, 2018; Cawley et al., 2010), education (Brunello et al., 2013; Clark and Royer, 2013), economic development (Ruhm, 2005), unemployment (Deb et al., 2011), and air pollution (Deschenes et al., 2020) to explain obesity trends. However, these factors alone cannot fully account for the continued rise in obesity rates over the past decade, particularly in developing countries.

We identify short videos as a previously overlooked but important determinant of obesity. As a rapidly growing form of digital entertainment, short video users are nearly 40 % of the global population. Understanding the causal impact of short video usage on obesity is crucial to explaining the recent surge in obesity rates. To the best of our knowledge, this is the first study to establish a causal link between short videos and obesity, offering a novel perspective for understanding and potentially mitigating the obesity epidemic.

To investigate this relationship, we use two years of micro-survey data from the China Family Panel Studies (CFPS) for 2020 and 2022. There are two key reasons for selecting China as the study sample. First, China's short video platforms have experienced rapid growth, with over one billion users, making it an ideal context for examining the impact of short video consumption. Second, as the world's largest developing country with the highest obesity rate, China provides valuable insights into the rising obesity trends in developing countries.

First, we examine the effect of short videos on body weight, including their impact on BMI and the probability of obesity by using fixed effect panel regression model. Our findings indicate that short videos contribute to an increase in BMI by 0.12 and the likelihood of obesity by 2.2 %. We estimate that the negative health impact of short videos costs

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China approximately CNY 7.53 billion one year. It can explain about 27 % of the growth in China's obese population from 2020 to 2022, suggesting that it has become a major new driver of obesity. We confirm the robustness of our results through various checks, including changing variables, adding additional fixed effects and pre-trend controls, and using Propensity Score Matching (PSM) and Instrument Variable (IV) method.

Next, we explore the mechanisms underlying this effect. We find that short videos are addictive, leading to increased screen time on mobile devices, particularly for entertainment purposes, which displaces time that could otherwise be spent on physical activity. This addiction appears to negate the positive effects of weight-loss information channel. Additionally, we observe no significant effects from other potential mechanisms, such as mental health, sleep quality, or health-related habits. This may be due to the offsetting nature of both positive and negative influences.

Finally, we conduct a heterogeneity analysis to understand the groups most affected by short videos. Our results show that men, people with lower levels of education or aged over 35, and urban residents are more susceptible to the negative impact of short videos on body weight. This suggests that short video addiction is most prevalent among these demographic groups, and that education could play a role in mitigating the adverse health effects of short videos.

This study contributes to the literature on the economic determinants of obesity by incorporating the role of short videos. Previous studies have predominantly examined factors such as food prices (Anderson and Matsa, 2011; Cawley, 2015; Christian and Rashad, 2009; Courtemanche and Carden, 2011; Currie et al., 2010), income (Akee et al., 2013, 2018; Cawley et al., 2010), education (Brunello et al., 2013; Clark and Royer, 2013), economic development (Ruhm, 2005), unemployment (Deb et al., 2011), and air pollution (Deschenes et al., 2020) to explain obesity trends. But they these factors fall short in explaining the rapid rise in obesity rates in developing countries over the past decade. We identify short videos as a novel and significant determinant of obesity to try to explain it.

Moreover, our findings contrast with those of Huang et al. (2025), who show that television viewing increases obesity but does not significantly reduce time spent on physical activity. But in our case, short videos crowd out exercise time through addiction-driven engagement. This indicates that short videos possess distinct and more potent addictive properties compared to traditional video formats, thereby exerting a stronger influence on obesity outcomes.

Our study advances this literature by distinguishing short videos from general Internet usage and explicitly investigating their causal impact on obesity. The impact of Internet access on obesity remains mixed. Some studies suggest that Internet access contributes to obesity through increased sedentary behavior (Aghasi et al., 2020; DiNardi et al., 2019), while others argue that it may reduce obesity by facilitating access to health-related information (Chen and Liu, 2022). Through mechanism analysis, we consider both potential positive and negative pathways. We find that short video exposure significantly increases obesity via the addiction channel. In contrast to earlier findings that the information channel can facilitate weight loss, our results suggest that this channel becomes ineffective in the context of short videos due to their highly addictive nature—resulting in a net negative impact on body weight. This distinction underscores the unique behavioral dynamics associated with short video platforms, which differ fundamentally from broader Internet use.

We extend the growing research on short video platforms by providing robust empirical evidence of their broader impact. Much of the existing literature focuses on how short video platforms boost profitability by enhancing user stickiness (Huang et al., 2025; Xiao et al., 2025), increasing viewing time (Zhang et al., 2024), stimulating consumer spending (Meng et al., 2024; Xiao et al., 2023), or improving the attractiveness of advertising (Cheung et al., 2022; Jin and Ryu, 2020; Masuda et al., 2022).

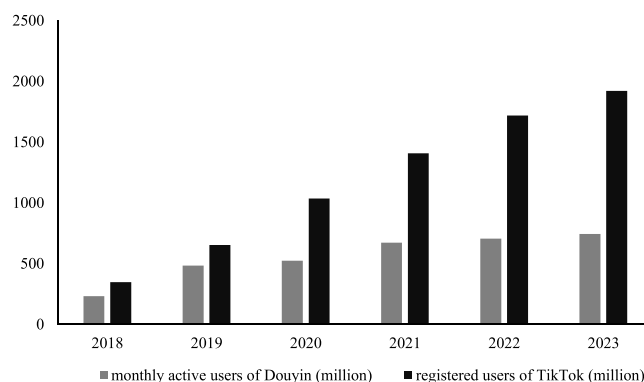


Fig. 1. Number of short videos users. This figure plots the monthly active users of Douyin and the registered users of TikTok from 2018 to 2023.

However, the public health implications of these platforms, particularly their effects on obesity, remain largely unexplored. By linking short video usage to obesity, we offer new insights into the consequences of these platforms beyond their commercial functions. Moreover, while previous research often relies on case studies (Zhang et al., 2024) or questionnaire surveys (Cheung et al., 2022; Jin and Ryu, 2020; Masuda et al., 2022) with small, non-random samples, our study employs large-scale, randomized micro-survey data from China, yielding more scientifically rigorous and generalizable results.

Our paper also contributes to the literature on the welfare effects of social media. Previous studies on social media have primarily focused on its psychological effects, such as reducing happiness (Allcott et al., 2022; Aridor et al., 2024) and increasing depression and anxiety. However, the link between social media and obesity, a physical health concern, remains largely unexplored. In addition, the existing literature has mainly studied Facebook (Allcott et al., 2022; Aridor et al., 2024), and few studies have focused on the role of short videos. Our study addresses this significant gap in the literature by examining the influence of short videos on obesity.

In summary, our research fills a critical gap by investigating the physical health effects of social media—specifically, the impact of short videos on obesity—and contributes to the broader debate on the causes of obesity. Utilizing micro-level data from China, the country with the largest short video user base, our findings identify short video usage as a new external determinant of health and offers a novel lens through which to assess the welfare effects of short video platforms.

The remainder of this paper is structured as follows. Section 2 introduces the background of short videos. Section 3 outlines the model and data. Section 4 presents the empirical results, including the baseline results and robustness checks. Section 5 discusses the mechanism. Section 6 presents the heterogeneity analysis. Section 7 does cost-benefit analysis. Section 8 concludes.

2. Background

Short videos have rapidly become a global phenomenon, as shown in Fig. 1. In China, the number of monthly active users of Douyin (the Chinese version of TikTok) grew from 18.5 million in 2017–743 million in 2023, with the total user base exceeding 1 billion in 2023.¹ TikTok, which launched its international version in late 2017, has also experienced exponential growth, with the number of registered users increasing from 347.1 million in 2018–1.92 billion in 2023. TikTok is now the most downloaded app in over 40 countries and is available in more than 150 markets and 35 languages. The widespread popularity of short videos has sparked various discussions, including concerns about

¹ The data is from QuestMobile

addiction. According to surveys, TikTok users spent an average of 850 min (14.1 h) per month on the app in 2023, up from 442.9 min (7.38 h) in 2019.²

Short video platforms are not only entertainment media but also function as social media networks (Aridor et al., 2024). Short videos platforms are different from traditional social media. While traditional social media primarily emphasizes user interaction and information sharing, short videos platforms rely heavily on sophisticated algorithms that track user behaviors—such as browsing, clicking, and engagement patterns—to deliver highly personalized content (Tian et al., 2022). This algorithm-driven curation creates an immersive experience that encourages prolonged viewing and heightens the risk of user addiction (Liao, 2024), often causing users to lose track of their time (Allcott et al., 2022; Zhang et al., 2024). Moreover, short videos, which are typically brief and engaging, promote rapid consumption and immediate entertainment (Tellis et al., 2019), in contrast to traditional social media's focus on fostering social connections (Lee et al., 2014).

Despite these unique characteristics, most existing research on social media has concentrated on its psychological impacts, such as reducing happiness and increasing depression and anxiety (Allcott et al., 2020; Braghieri et al., 2022; Kross et al., 2021; Yu et al., 2024), while relatively little attention has been paid to its effects on physical health, particularly obesity. This study aims to fill this gap by examining the influence of short video usage on body weight and obesity.

The effects of short videos on obesity are theoretically ambiguous and remain empirically underexplored. On one hand, excessive time spent watching short videos may crowd out physical activity, contributing to weight gain and obesity. On the other hand, short videos can serve as a medium for disseminating health and fitness information—such as workout tutorials or dietary advice—potentially encouraging healthier lifestyles and reducing obesity (Aghasi et al., 2020; Chen and Liu, 2022; DiNardi et al., 2019; Huang et al., 2025).

However, short videos differ fundamentally from traditional Internet or television content in one key respect: their algorithm-driven, highly personalized recommendation systems. These systems significantly increase user engagement and addictive behavior (Allcott et al., 2020, 2022; Aridor et al., 2024; Braghieri et al., 2022), which may amplify their negative health consequences. This behavioral distinction suggests that short video platforms could have a more pronounced effect on obesity than earlier forms of media.

In the following sections, we provide empirical evidence to identify the causal relationship between short video usage and obesity.

3. Model and data

3.1. Model

To estimate the causal impact of short videos on body mass, we utilize individual-level data from the China Family Panel Studies (CFPS) for 2020 and 2022. We use fixed effect panel regression model, and the specific model is as follow:

$$Y_{it} = \beta_1 \text{short videos}_{it} + \text{Control}_X \beta_X + \theta_i + \sigma_t + \varepsilon_{it} \quad (1)$$

Where, Y_{it} represents the body mass indicator for individual i in year t , which includes both BMI and an obesity indicator (Deschenes et al., 2020). BMI is computed from the individual's height and weight, while the obesity index is a binary variable indicating whether BMI exceeds 28 (Bei-Fan, 2003). The key explanatory variable, *short video*, is a dummy variable denoting whether the individual watches short videos. Control variables include both individual-level characteristics (age, age squared, gender, household registration, education, and medical insurance) and family-level variables (logarithm of household income and household

size). The model controls for individual fixed effects θ_i and year fixed effects σ_t , and standard errors are clustered at the province level to account for potential spatial correlation.

3.2. Data

Our primary data source is the China Family Panel Studies (CFPS), which provides comprehensive longitudinal information on social, economic, demographic, educational, and health variables across China (Xie and Hu, 2014). CFPS is a collection of microdata from a biennial follow-up survey of Chinese households and individuals. The sample selection is based on the principle of random sampling and is nationally representative (Fan et al., 2021).

The CFPS contains detailed individual-level data, including height and weight, which allows us to compute both BMI and the obesity index. BMI is calculated according to the formula for calculating BMI, which is the weight divided by the square of height. Obesity indicators we use the Chinese standard, that is, a BMI greater than 28 is considered obese (Bei-Fan, 2003). In Fig. 2, we plot the density of our BMI index. We find that the BMI index follows a normal distribution within the normal range, indicating the reliability of our sample.

Importantly, questions about short videos were included in the 2020 and 2022 waves of the survey, which limits our analysis to these two years. The data shows that the proportion of people watching short videos has increased from 81.91 % in 2020–89.33 %.

In addition, detailed individual and household surveys of the CFPS provide enough individual- and household-level control variables to prevent the omitted variable problem. The detailed definitions of the variables are provided in Appendix Table A1. We focus on individuals aged 16 years and older. Because the individuals below 16 are impossible to have the mobile phone to watch short videos in China and their bodies have not fully developed yet. We then exclude respondents with missing values for body mass index (BMI), height, or weight, as well as those lacking information on short video usage or key control variables. After these exclusions, the final sample includes 8194 individuals. The summary statistics are presented in Table 1.

4. Empirical results

4.1. Baseline

Table 2 presents the results of estimating the impact of short videos on body mass. The dependent variables are BMI (column 1) and the obesity indicator (column 2). We find that short videos significantly increase the BMI at the 5 % level and raises the probability of obesity at

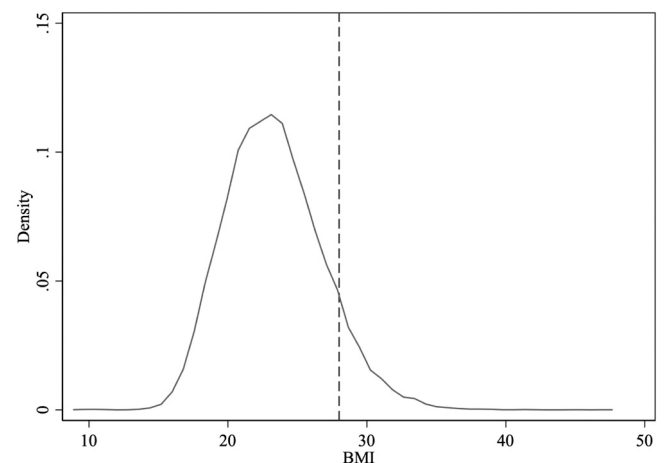


Fig. 2. Density of the BMI index. This figure plots the density of the BMI index. The kernel function is epanechnikov.

² See <https://blog.hootsuite.com/tiktok-stats/>, accessed January 23, 2025.

Table 1
Summary statistics.

Variables	Obs	Mean	Std. Dev.	Min	Max
BMI	16,388	23.345	3.483	9.306	47.241
Obesity	16,388	0.096	0.295	0	1
Short videos	16,388	0.856	0.351	0	1
Age	16,388	39.083	13.259	16	88
Age ²	16,388	1703.298	1133.051	256	7744
Gender	16,388	0.512	0.5	0	1
Education	16,388	3.579	2.054	0	9
Urban	16,388	0.61	0.488	0	1
Medical insurance	16,388	0.918	0.274	0	1
ln(family income)	16,388	11.468	0.823	2.303	16.248
Family size	16,388	4.047	1.934	1	16

Table 2
The effects of the short videos on body mass.

Variables	(1)	(2)
	BMI	Obesity
Short videos	0.117** (0.055)	0.022*** (0.007)
Controls	Yes	Yes
Individual FE	Yes	Yes
Year FE	Yes	Yes
Baseline Mean	23.079	0.0735
Observations	16,388	16,388
Adjusted R-squared	0.866	0.667

This table reports the effects of the short videos on body mass. The dependent variables are BMI in column (1), and an obesity indicator in column (2). All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

the 1 % level. Specifically, we find that watching short videos increases BMI by 0.12 and raises the probability of obesity by 2.2 % points. Among individuals who did not watch short videos in 2020—the baseline mean BMI and obesity rate are 23.079 and 7.35 %, respectively. These baseline values provide a reference point for assessing the magnitude of the estimated effects, indicating that short video watching increases BMI by approximately 0.52 % and the probability of obesity by about 29.9 % relative to the baseline.

Chen and Liu (2022) find that broadband access would reduce the obesity rate in China, which is different from our conclusion because we studied a more specific Internet application, short videos. This indicates that different Internet applications may lead to different effects, providing an important explanation for the different effects of Internet access on obesity.

In conclusion, we have demonstrated that short videos are a novel factor in obesity. The government should take corresponding actions to prevent the harm of short videos.

4.2. Robustness test

We conduct a series of additional analyses to check the robustness of our results in Table 3, including changing measure of variables, adding fixed effects and pre-trend controls, eliminating the impact of the epidemic and some abnormal samples. Then we also do some endogeneity tests in Table 4, including Propensity Score Matching (PSM) method and Instrument variable (IV) method.

4.2.1. Change the variable

We first replace the core explanatory variable and the dependent variables with alternative measures in Panel A in Table 3. Columns (1) and (2) shows that using a dummy variable for short videos daily still results in significantly positive effects on body mass. In Columns (3) and (4), we substitute height and weight for the obesity indicator. The results

confirm that short videos increase weight, but do not significantly affect height. This finding is consistent with the idea that height is less likely to change during adulthood. Overall, all results demonstrate the robustness of the baseline regression.

4.2.2. Control more variables

We expand our model by adding more fixed effects in Panel B and pre-trend controls in Panel C to address the potential omitted variable bias in Table 3. In Panel B, we add month fixed effects in Columns (1) and (2), county-year fixed effects in Columns (3) and (4), and province-year fixed effects in Columns (5) and (6) to control temporal and spatial omitted variables. The results show that even after adding more fixed effects, the core results are the same: short videos have a significantly positive effect on body mass.

Next, we incorporate pre-trend controls at the county level to account for any pre-existing trends in obesity in Panel C in Table 3. After controlling for the linear, quadratic, and cubic terms of the county-level pre-trends, the results don't change. In all, these findings remain robust, confirming that the effects of short videos on body mass are not driven by pre-existing trends.

In Appendix Table A2, we control overall online duration on mobile devices in the regression as a horse-race. Even after controlling for total mobile device use, short video viewing still has an impact on body weight. This finding underscores that the effect of short video consumption on body weight is not merely due to overall mobile usage, highlighting the distinct role of short videos.

In Appendix Table A3, we address the potential two-stage selection issue by excluding the effect of mobile phone ownership. Since watching short videos requires owning a smartphone, it is possible that the observed effect of short video use might partly capture the effect of mobile phone ownership itself. To test this, we firstly use mobile phone ownership as the core explained variable and we don't find significant impact on BMI and obesity. Secondly, after controlling the mobile phone ownership, the effect of short video use remains statistically significant. Thus, this suggests that our findings are not driven by this selection bias.

4.2.3. Eliminate samples

We eliminate samples to prevent abnormal samples from interfering with our results. To isolate the impact of the COVID-19 pandemic, which may have influenced individual health outcomes, we re-estimate the model separately for 2020 and 2022. The results in Panels E of Table 3 show that the coefficients remain significantly positive in both years, suggesting that the observed effects are not driven by pandemic-related factors. Thus, after eliminating the impact of the pandemic, the results are still robust.

In Panel E of Table 3, we exclude the samples with gender change and household registration change, which is not very normal thing. After excluding these sample, the results are still significantly positive.

4.2.4. Propensity score matching (PSM)

Since the explanatory variable is whether an individual watches short videos, there may be other factors influencing an individual's self-selection of whether to watch short videos. Given the potential endogeneity of short videos, we apply the Propensity Score Matching (PSM) method to reduce selection bias. We use three matching algorithms: k-nearest neighbors, radius matching, and kernel matching. As shown in Fig. 3, the standardized bias for all covariates is reduced to below 10 % after matching, indicating a successful balancing. The results in Panel A of Table 4 demonstrate that, regardless of the matching method, short videos still significantly increase BMI and the likelihood of obesity.

4.2.5. Instrument variable (IV)

Although we use PSM method to solve potential endogenous problem, the PSM method only can solve self-selection problem caused by the observed. As for the problem of unobservable factors, we use IV method to solve it.

Table 3
Robustness checks.

Panel A: Change the variables						
	(1)	(2)	(3)	(4)		
	Change the explanatory variable		Change the explained variable			
VARIABLES	BMI	Obesity	Weight	Height		
Short videos			0.731** (0.286)	0.043 (0.072)		
Watch short videos everyday	0.106* (0.052)	0.013* (0.007)				
Controls	Yes	Yes	Yes	Yes		
Individual FE	Yes	Yes	Yes	Yes		
Year FE	Yes	Yes	Yes	Yes		
Observations	16,388	16,388	16,388	16,388		
Adjusted R-squared	0.866	0.667	0.920	0.959		
Panel B: Add fixed effects						
	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	BMI	Obesity	BMI	Obesity	BMI	Obesity
Short videos	0.113** (0.054)	0.022*** (0.007)	0.138** (0.058)	0.018** (0.008)	0.138** (0.058)	0.018** (0.008)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
County × Year FE	No	No	Yes	Yes	Yes	Yes
Province × Year FE	No	No	No	No	Yes	Yes
Observations	16,388	16,388	15,286	15,286	15,286	15,286
Adjusted R-squared	0.866	0.667	0.869	0.665	0.868	0.662
Panel C: Add pre-trend controls						
	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	BMI	Obesity	BMI	Obesity	BMI	Obesity
Short videos	0.120** (0.054)	0.022*** (0.007)	0.120** (0.054)	0.022*** (0.007)	0.120** (0.054)	0.022*** (0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County-level pre-trend × <i>T</i>	Yes	Yes	Yes	Yes	Yes	Yes
County-level pre-trend × <i>T</i> ²	No	No	Yes	Yes	Yes	Yes
County-level pre-trend × <i>T</i> ³	No	No	No	No	Yes	Yes
Observations	16,152	16,152	16,152	16,152	16,152	16,152
Adjusted R-squared	0.867	0.670	0.867	0.670	0.867	0.670
Panel D: Eliminate the impact of the epidemic						
	(1)	(2)	(3)	(4)		
	2020		2022			
VARIABLES	BMI	Obesity	Weight	Height		
Short videos	0.266*** (0.086)	0.024*** (0.006)	0.272** (0.109)	0.024* (0.013)		
Controls	Yes	Yes	Yes	Yes		
Observations	8194	8194	8194	8194		
Adjusted R-squared	0.124	0.0152	0.104	0.0128		
Panel E: Eliminate some abnormal samples						
	(1)	(2)	(3)	(4)	(5)	(6)
	Gender		Household registration		Both	
VARIABLES	BMI	Obesity	BMI	Obesity	BMI	Obesity
Short videos	0.115** (0.055)	0.022*** (0.007)	0.133** (0.060)	0.021** (0.008)	0.131** (0.061)	0.021** (0.008)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,376	16,376	15,394	15,394	15,382	15,382
Adjusted R-squared	0.866	0.667	0.867	0.670	0.867	0.669

This table reports the results of robustness checks. In Panel A, we change the explanatory variable and the explained variable. In Columns (1) and (2), we use the variable of whether the person watches short videos everyday as the core explanatory variable. In Columns (3) and (4), we use height and weight as the explained variables, respectively. In Panel B, we add more fixed effects, including month fixed effect, county-year fixed effect, and province-year fixed effect. In Panel C, we add pre-trend controls, including the primary, quadratic and cubic term of county-level pre-trend. In Panel D, we eliminate the impact of the epidemic. We use one year data to present the results. In Panel E, we eliminate some abnormal samples. In Columns (1) and (2), we exclude the samples with gender changes. In Columns (3) and (4), we exclude the samples with household registration changes. In Columns (5) and (6), the two changes sample both are excluded. All the results control the individual and year fixed effects except the Panel D. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Based on [Manacorda and Tesei \(2020\)](#), we use lightning frequency as the IV. Specifically, we calculate the average lightning frequency at the provincial level in China from 1995 to 2014 based on the total global flash frequency of lightning monitored by the Optical Transient Detector (OTD) on the MicroLab-1 satellite. To prevent absorption by the fixed

effect of the year, we multiplied the average value by the changes in short video viewing at the household level. This instrumental variable satisfies the correlation. The higher the frequency of lightning, the higher the travel risk and the easier it is to watch short videos at home for entertainment. Secondly, it meets the exogeneity requirement. The

Table 4
Endogeneity tests.

Panel A: PSM method						
	(1)	(2)	(3)	(4)	(5)	(6)
	K-Nearest neighbors		Radius		Kernel	
VARIABLES	BMI	Obesity	BMI	Obesity	BMI	Obesity
Short videos	0.117** (0.055)	0.022*** (0.007)	0.117** (0.055)	0.022*** (0.007)	0.117** (0.055)	0.022*** (0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,382	16,382	16,382	16,382	16,382	16,382
Adjusted R-squared	0.866	0.667	0.866	0.667	0.866	0.667
Panel B: IV method						
	(1)	(2)	(3)			
	First stage	Second stage				
VARIABLES	Short videos	BMI	Obesity			
Short videos		0.628** (0.310)	0.046* (0.025)			
IV	0.042*** (0.004)					
Cragg-Donald Wald F statistic	557.539					
Kleibergen-Paap rk Wald F statistic	138.583					
Controls	Yes	Yes		Yes		Yes
Individual FE	Yes	Yes		Yes		Yes
Year FE	Yes	Yes		Yes		Yes
Observations	13,524	13,524				13,524
Adjusted R-squared	0.337	0.0008				0.0005

This table reports the results of endogeneity tests. In Panels A and B, we use Propensity Score Matching (PSM) method and Instrument variable (IV) method, respectively. In Panel A, we use different match method including k-nearest neighbors matching, radius matching and kernel matching. Standard errors are cluster at the province-level in parentheses. In Panel B, we show the results of the first stage in Column (1), and the second stage in Columns (2) and (3). Standard errors are cluster at the household-level in parentheses. All the results control the individual and year fixed effects. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

local historical lightning frequency has no direct relationship with an individual's obesity.

The results are shown in Panel B of Table 4. In Column (1), the IV has a significant positive relationship with watching short videos. The Cragg-Donald Wald F statistic and Kleibergen-Paap rk Wald F statistic are 557.539 and 138.583 respectively, which are enough big to overcome the weak IV identify. In Columns (2) and (3), the results are still significantly positive. In all, after solving the unobserved endogenous problem, the results are still robust and credible.

5. Mechanism analysis

We explore the mechanisms through which short videos affect body mass. One possible channel is obvious that short videos take place of physical exercise time. Short videos have addictive nature (Allcott et al., 2020, 2022; Aridor et al., 2024; Braghieri et al., 2022). Short videos can reduce the time available for physical exercise, thereby leading to weight gain.

However, another possibility is that fitness-related content on short videos could encourage viewers to engage in exercise, potentially counteracting negative health effects. Since February 18, 2022, Liu Yonghong has led audiences in fat-burning aerobics through live broadcasts on short video platforms, significantly influencing exercise habits via these videos. This form of exercise has gained widespread popularity, attracting tens of millions of new followers each day. Within less than a month, the self-created exercise videos had been viewed over a billion times. According to surveys, by 2022, nearly 30 million adults had participated in Liu Yonghong's exercise routines more than ten times. This presents the competing hypothesis that short videos may promote physical exercise.

Although our results demonstrate that short videos increase body weight, it remains an important avenue for future research to explore whether short videos might also encourage or hinder exercise engagement. To disentangle these two competing mechanisms, we conduct

further analysis.

5.1. The addictive effects of short videos

First, we examine the addiction of short video in Table 5. In Panel A, we take the duration of watching short videos every day as the explanatory variable.³ We find that if one person watches more 1 h for short videos, the BMI index will increase by 0.06 and the probability of obesity will increase by 1.2 %. Therefore, controlling the duration of watch short videos is necessary for our health.

Second, we show how this addiction changes our opinion of using the Internet. In Panel B, we use the individual's perception of the importance of the Internet to certain aspects as the explained variable (Chen, 2025). We find short videos will make people think the Internet is important to entertainment and life, instead of socialization, study and work. Therefore, short videos can make people addicted to the pleasures of the Internet.

These findings suggest that short videos are highly addictive. We then proceed to explore whether the impact of short videos on exercise is negative or positive.

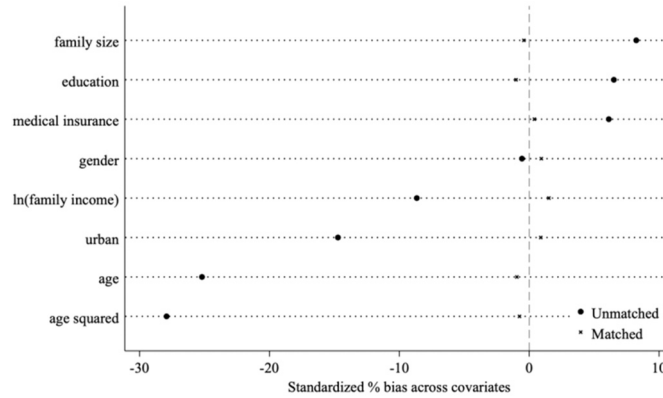
5.2. The negative effects of short videos on exercise

Next, we examine the negative effects of short videos on exercise, as shown in Panel A of Table 6. We regress several exercise-related variables as dependent variables: whether an individual exercises (column 1), the duration of exercise per session (column 2), and the frequency of exercise (column 3).

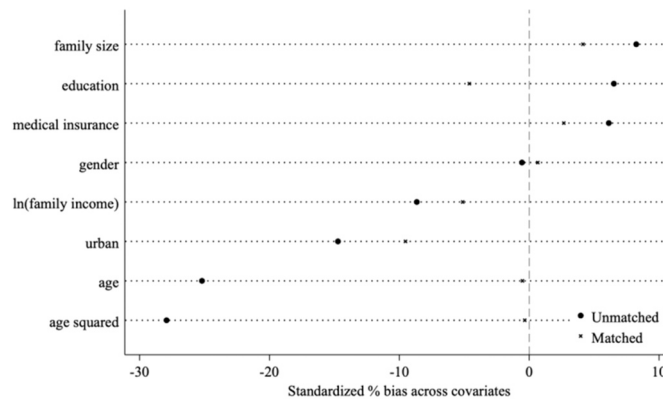
Our results indicate that short videos significantly reduce the likelihood of exercising, the duration of exercise per session, and exercise

³ The short videos duration is roughly calculated by the average of the time spent on mobile devices difference between watching short videos and not watching short videos in the province level for watching short videos people.

Panel A: K-Nearest neighbors matching



Panel B: Radius matching



Panel C: Kernel matching

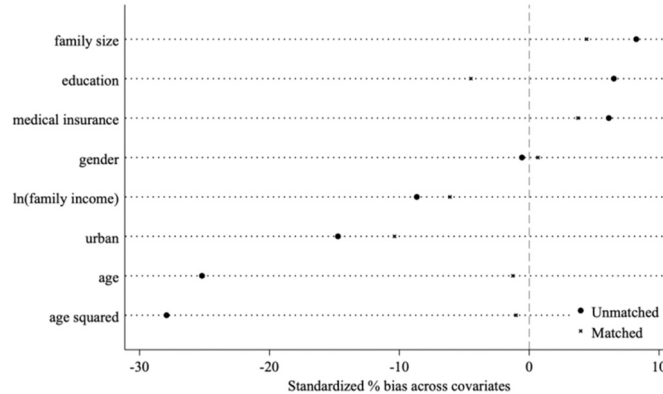


Fig. 3. Balance test. This figure plots the balance test of PSM. Dots are results before matching, and forks are results after matching.

frequency, all at the 1 % significance level. These findings provide strong evidence that short videos have detrimental effects on exercise habits. This explains the important reason why short videos can cause obesity.

This result is different from the findings of traditional media TV. Huang et al. (2025) find that watching TV does not affect obesity. It proves the uniqueness of short videos, which make people's addictive through the recommendation algorithm and made it easier to replace exercise.

5.3. The positive effects of short videos on exercise

Then, we test the competing hypothesis that short videos may positively affect exercise by information channel. Liu Yonghong is the first individual to popularize exercise through live broadcasts on short video platforms. Therefore, we use the date of his first live broadcast, February 18, 2022, as an exogenous shock to examine the impact of short videos on exercise. Because data including the month of survey, and the survey months are mainly in the second half of the year. We draw the timeline in Fig. 4 to show the time of shock and survey. Therefore, we employ a difference-in-differences (DID) approach to estimate the effect, as follows:

Table 5

The additive effects of short videos.

Panel A: Study the influence of the length of short videos time		
	(1)	(2)
VARIABLES	BMI	Obesity
Short videos duration	0.001** (0.000)	0.0002*** (0.000)
Controls	Yes	Yes
Individual FE	Yes	Yes
Year FE	Yes	Yes
Observations	16,388	16,388
Adjusted R-squared	0.866	0.667

Panel B: Study the impact of short videos on people's concepts of using the Internet					
	(1)	(2)	(3)	(4)	(5)
VARIABLES	Socialization	Entertainment	Study	Work	Life
Short videos	0.046 (0.039)	0.144** (0.053)	0.042 (0.072)	0.041 (0.053)	0.164*** (0.055)
Controls	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	16,382	16,378	16,378	16,338	16,374
Adjusted R-squared	0.322	0.375	0.369	0.390	0.387

This table reports the additive effects of short videos. In Panel A, we study the influence of the length of short videos time. In Panel B, we study the impact of short videos on people's concepts of using the Internet. All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Table 6

The effect of short videos on exercise.

Panel A: The negative effect of short videos on exercise			
VARIABLES	(1)	(2)	(3)
	Exercise	Exercise duration	Exercise frequency
Short videos	-0.246***	-12.985***	-0.926***
	(0.030)	(1.206)	(0.109)
Controls	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	14,298	14,298	14,298
Adjusted R-squared	0.457	0.371	0.470
Panel B: Competing hypothesis			
VARIABLES	(1)	(2)	(3)
	Exercise	Exercise duration	Exercise frequency
Short videos	-0.950***	-14.006***	-0.251***
	(0.121)	(1.720)	(0.034)
Short videos*post	0.072	3.051	0.014
	(0.116)	(3.120)	(0.026)
Controls	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	14,298	14,298	14,298
Adjusted R-squared	0.470	0.371	0.457

This table reports the effect of short videos on exercise. The dependent variables are whether exercise or not in column (1), exercise duration for one time in column (2) and exercise frequency in column (3). In Panel A, we examine the bad effect of short videos on exercise. In Panel B, we check the competing hypothesis. All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

$$Exercise_{it} = \beta_1 short\ videos_{it} \times Post + \beta_2 short\ videos_{it} + Control_x \beta_x + \theta_i + \sigma_t + \varepsilon_{ict} \quad (2)$$

In this model, the explained variables *Exercise* include whether exercise or no, exercise duration for one time, exercise frequency. The variable *Post* is a dummy variable, coded as 1 for the time after February

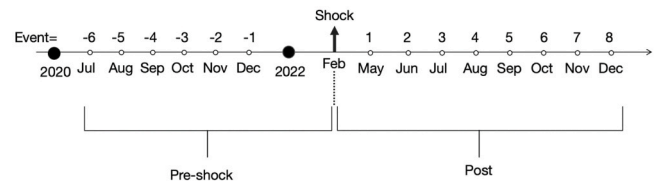


Fig. 4. Timeline of shock. This figure plots the timeline of shock. The write dots represent the survey month, and the figure highlights the time of shock.

2022 and 0 otherwise. The coefficient of primary interest, β_1 , reflects the potential positive effect of short videos on exercise. If short videos indeed encourage exercise, we expect β_1 to be significantly positive. All other settings are the same as in the baseline regression.

The estimated results are presented in Panel B of Table 6. We find that the coefficients for the interaction terms are not significant, and the coefficient for the short video variable remains significantly negative. This suggests that short videos do not have a positive effect on exercise.

Furthermore, we conduct an event study to further examine this hypothesis. The regression equation is as follows:

$$Exercise_{it} = \beta_0 + \sum_{j=-1} \beta_j \times 1(Event_{it} = j) + Control_x \beta_x + \theta_i + \sigma_t + \varepsilon_{ict} \quad (3)$$

In this model, $Event_{it}$ represents the time minus the Liu's first live broadcast time measured by month. The dummy variable $1(Event_{it} = j)$ indicates whether the observation corresponds to the current period. For the value of event, we set it from -6 – 8 in sequence according to the distance from shock. As shown in Fig. 4, $t = -6, -5, \dots, -1$ represent six months prior to the shock (from July to December 2021), and $t = 1, 2, \dots, 8$ correspond to the post-shock months (from May to December 2022). For individuals who do not watch short videos, the variable $1(Event_{it} = j)$ is set to zero for all time periods. We exclude the $j = -1$ as the baseline to avoid multicollinearity.

The results are shown in Fig. 5. In Panels A, B, and C, we regress the dependent variables—whether the individual exercises, exercise duration per session, and exercise frequency, respectively—on the event time dummies. We find that none of the coefficients are statistically significant, both before and after Liu's first live broadcast. This checks the parallel trends test, further reinforcing the conclusion that short videos have no significantly positive effect on exercise.

In summary, we identify exercise-related mechanisms that explain why short videos may contribute to weight gain. The addictive nature of short videos leads to a reduction in the time available for physical activity. We also reject the competing hypothesis that short videos promote exercise, concluding that short videos have no significant positive impact on exercise. This shows that short videos do lead to the failure of Internet information channels through the addiction mechanism, so how to prevent addiction has become the top priority.

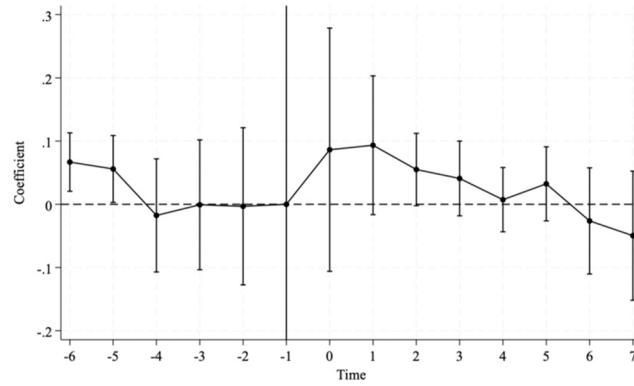
5.4. Other mechanisms

We also examine additional mechanisms, including mental health, sleep quality, and health-related habits, as reported in Table 7.

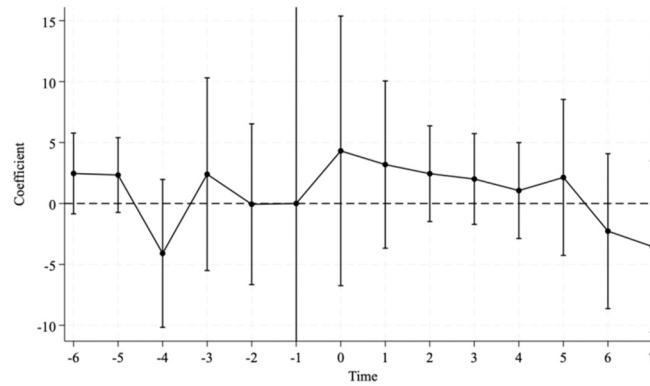
In Panel A, we investigate the impact on mental health, focusing on depression and subjective well-being. Depression is measured using scores from the 20-item and 8-item versions of the Center for Epidemiologic Studies Depression Scale (CES-D), with higher scores indicating more severe depressive symptoms. Subjective well-being is self-reported on an integer scale from 1 to 10, where higher values reflect greater happiness. The results indicate that short-form video consumption has no statistically significant effect on either depression or well-being, suggesting no substantial impact on mental health.

A plausible explanation for these null findings is the offsetting interplay between mood regulation and social substitution. On one hand, short-form videos can serve as a low-cost source of entertainment

Panel A: Exercise



Panel B: Exercise duration



Panel C: Exercise frequency

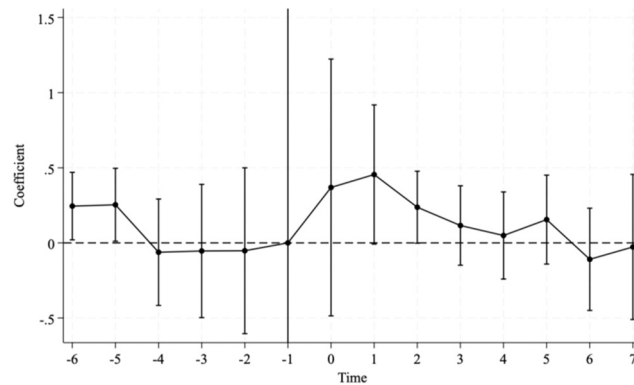


Fig. 5. Event study results. This figure plots the event study results of the effects of the shock on exercise. The dependent variables are whether exercise or not, exercise duration for one time and exercise frequency in Panel A, B and C, respectively. Dots are coefficients and segment lengths are 95 % confidence intervals.

and temporary distraction, potentially generating positive affect and alleviating stress. On the other hand, excessive immersion may displace offline social interactions and foster feelings of social isolation or upward social comparison, which can undermine psychological well-being. The concurrent presence of these opposing forces may result in a net effect that is statistically indistinguishable from zero.

In Panel B, we analyze the effects on sleep, using two measures: poor sleep frequency and sleep duration. Poor sleep is a categorical variable constructed based on the frequency of sleep difficulties, coded as 0 for "rarely or never," 1 for "sometimes," 3 for "often," and 4 for "most of the time." Sleep duration refers to the average number of hours slept per night, excluding naps. Again, we find no significant effects.

This lack of significance may be attributed to the countervailing mechanisms of cognitive relaxation and cognitive arousal. For some users, viewing short-form videos before bed may function as a form of

psychological detachment from daily stresses, thereby facilitating sleep onset. For others, the blue light emitted from devices and the stimulating nature of the content can suppress melatonin production, delay sleep, and reduce sleep quality. The heterogeneity in user routines and content consumption likely leads to these effects canceling each other out in the aggregate.

In Panel C, we explore the influence on dietary patterns. We capture nutritional intake using survey items on whether respondents consumed meat and vegetables in the past week, as proxies for protein and vegetable consumption. Similarly, we detect no significant associations.

The null result here may stem from the competing influences of health knowledge dissemination and unhealthy behavioral priming. The platform hosts a substantial volume of content promoting healthy recipes and nutritional literacy, which could encourage higher consumption of proteins and vegetables. Simultaneously, it also features

Table 7
Other mechanisms.

Panel A: Mental health.				
	(1)	(2)	(3)	
VARIABLES	Cesd20	Cesd8	Happiness	
Short videos	-0.303 (0.227)	-0.157 (0.114)	-0.031 (0.083)	
Controls	Yes	Yes	Yes	
Individual FE	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	
Observations	16,338	16,338	16,384	
Adjusted R-squared	0.517	0.517	0.515	
Panel B: Sleep.				
	(1)	(2)		
VARIABLES	Bad sleep	Sleep time		
Short videos	-0.017 (0.029)	0.055 (0.053)		
Controls	Yes	Yes		
Individual FE	Yes	Yes		
Year FE	Yes	Yes		
Observations	16,378	10,266		
Adjusted R-squared	0.420	0.435		
Panel C: Diet.				
	(1)	(2)		
VARIABLES	Protein	Vegetables		
Short videos	0.015 (0.015)	0.008 (0.006)		
Controls	Yes	Yes		
Individual FE	Yes	Yes		
Year FE	Yes	Yes		
Observations	16,386	16,386		
Adjusted R-squared	0.298	0.0411		
Panel D: Bad habits.				
	(1)	(2)	(3)	(4)
VARIABLES	Smoke last month	Smoke last week	Smoking amount	Wine
Short videos	-0.002 (0.006)	0.012 (0.102)	0.013 (0.008)	0.002 (0.011)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	16,386	16,388	11,568	16,384
Adjusted R-squared	0.861	0.846	0.664	0.536

This table reports the other mechanisms, including mental health in Panel A, sleep in Panel B, bad habits in Panel C, bad habits in Panel D. All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

'mukbang' (eating shows) and advertisements for high-calorie foods, which may trigger cravings and unhealthy eating behaviors. When these positive and negative influences coexist across the user population, the observed average effect on dietary intake can be neutral.

Finally, in Panel D, we assess the relationship with risky health behaviors, namely smoking and alcohol consumption. Smoking behavior is measured through three variables: whether the respondent smoked in the past month, whether they smoked in the past week, and the quantity of cigarettes smoked. Alcohol use is measured by whether the individual consumed alcohol at least three times in the past month. We still observe no significant effects.

This non-significance can be explained by the offsetting effects of health warnings and social normalization. Algorithmic systems on short-form video platforms may expose some users to public service announcements and graphic health warnings about the dangers of smoking and drinking, which could deter such behaviors. Conversely, the prevalent depiction of these substances in social and entertainment contexts may create a perception of social acceptability and even desirability, particularly among younger users. The simultaneous delivery of these conflicting sign.

In conclusion, the above-mentioned mechanisms are not significant because they are offset by both positive and negative influences. As for the exercise mechanism, we only found the negative influence. So short

videos mainly lead to obesity by reducing exercise.

6. Heterogeneity analysis

Table 8 presents the heterogeneity analysis of the effects of short videos on body mass. In Panel A, we examine the heterogeneity by gender. The results indicate that short videos have no significant effect on female body mass but have a significant positive effect on male body mass. This suggests that men are more likely to become addicted to short videos than women.

In Panel B, we investigate heterogeneity by age, dividing individuals

Table 8
Heterogeneity analysis.

Panel A: Heterogeneity by gender				
	(1) Male	(2) Female	(3) Male	(4) Female
VARIABLES	BMI	BMI	Obesity	Obesity
Short videos	0.239*** (0.084)	-0.009 (0.086)	0.035*** (0.012)	0.008 (0.009)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	8378	7998	8378	7998
Adjusted R-squared	0.854	0.871	0.669	0.656
Panel B: Heterogeneity by age				
	(1) < 35	(2) > =35	(3) < 35	(4) > =35
VARIABLES	BMI	BMI	Obesity	Obesity
Short videos	0.010 (0.113)	0.173** (0.074)	0.007 (0.011)	0.030** (0.011)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	7036	8756	7036	8756
Adjusted R-squared	0.849	0.868	0.649	0.675
Panel C: Heterogeneity by education				
	(1) High education	(2) Low education	(3) High education	(4) Low education
VARIABLES	BMI	BMI	Obesity	Obesity
Short videos	-0.024 (0.142)	0.137* (0.073)	0.007 (0.016)	0.031*** (0.008)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	3960	9346	3960	9346
Adjusted R-squared	0.855	0.864	0.648	0.657
Panel D: Heterogeneity by rural-urban				
	(1) Urban	(2) Rural	(3) Urban	(4) Rural
VARIABLES	BMI	BMI	Obesity	Obesity
Short videos	0.165* (0.093)	0.069 (0.153)	0.026** (0.012)	0.012 (0.010)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	9494	5900	9494	5900
Adjusted R-squared	0.863	0.871	0.685	0.643

This table reports the heterogeneity analysis of short videos on body mass. In Panel A, we illustrate the heterogeneity by gender. In Panel B, we try to find the heterogeneity by age. People whose age are over 65 are old, otherwise young. In Panel C, we find the heterogeneity by education. Individuals are divided according to median education level. Above the median is the high-education group and below the median is the low-education group. In Panel D, we illustrate the heterogeneity by rural-urban. The dependent variables are BMI in column (1) and (3), and an obesity indicator ($BMI \geq 28$) in column (2) and (4). All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

into two groups based on the median age: those aged above 35 and those below 35. We find that the negative effect of short videos on body mass is observed only among individuals aged above 35, while no significant effect is found among younger individuals. There are two possible reasons for this: The first is that after the age of 35, one has more time to enjoy life due to age discrimination and thus spends more time watching short videos. Because 35 is often the highest age required for job advertisements in China, people over 35 are subject to age discrimination in the job market (Lei and Chen, 2025). The second is that physical fitness declines, making one more susceptible to negative influences.

In Panel C, we explore heterogeneity by education level, categorizing individuals into high and low education groups based on the median education level. The results show that short videos have no significant effect on individuals with higher education but have a significant positive effect on individuals with lower education levels. This suggests that individuals with lower education levels are more likely to become addicted to short videos.

In Panel D, we analyze heterogeneity by rural-urban status. Individuals are divided into rural and urban groups based on household registration information. We find that short videos only have a significant positive effect on urban individuals, while no such effect is observed for rural individuals. This is likely due to limited access to the Internet in rural areas, which makes it more difficult for rural individuals to be influenced by short videos.

In summary, the heterogeneity analysis reveals that people aged over 35, men, individuals with lower education levels, and urban individuals are more susceptible to the negative effects of short videos. Therefore, strengthening the education of netizens' fitness awareness may be the way to solve the problem of weight gain caused by short videos

7. Cost-Benefit analysis

Previously, we have identified the negative impact of short video consumption on body weight, yet its broader economic implications remained unexplored. This section quantifies the economic losses associated with obesity induced by short video use, providing a reference for policymakers to assess the cost of addressing this emerging public health issue.

First, we estimate the number of individuals who became obese due to short video consumption. According to the coefficient in Column (2) of Table 2, short video use increases the probability of obesity by 2.2 %. Based on the *China Short Video Development Report (2024)* issued by the Development Research Center of the National Radio and Television Administration, the number of short video users in China rose from 873 million in 2020–1.012 billion in 2022. Thus, approximately 3.06 million individuals ($= 2.2\% \times [1.012 - 0.873]$ billion) became obese due to short video consumption during this period, implying an average annual increase of 1.53 million people.

According to the World Health Organization, China's adult obesity rate increased from 7.3 % in 2020–8.3 % in 2022, corresponding to an increase of 11.28 million individuals.⁴ Therefore, short video use can explain about 27 % of the growth in China's obese population from 2020 to 2022, suggesting that it has become a major new driver of obesity.

Next, we assess the economic consequences of obesity. Obesity significantly raises the incidence of diabetes, cardiovascular diseases (including heart disease and stroke), and cancers (Zhou et al., 2024). According to Qin and Pan (2016), obesity accounts for 5.29 % of total per capita healthcare expenditure related to these conditions. With per capita medical expenses of CNY 4656.7 in 2019, this implies an additional CNY 246 ($= 4656.7 \times 5.29\%$) in healthcare costs per person

attributable to obesity. Multiplying this by the 3.06 million obesity cases induced by short videos yields an aggregate annual cost of approximately CNY 7.53 billion ($= 246 \times 3.06$ million) in one year.

For comparison, Deschenes et al. (2020) estimate that a $1 \mu\text{g}/\text{m}^3$ increase in average PM2.5 concentration results in an economic cost of CNY 2.05 billion (USD 0.31 billion) through obesity-related health impacts. This suggests that the effect of short video consumption is equivalent to an increase of about $3 \mu\text{g}/\text{m}^3$ in PM2.5 exposure. These results highlight that obesity induced by short video use imposes substantial economic losses on the national economy, underscoring the need for targeted public health and regulatory policies to mitigate its impact.

8. Conclusion

Obesity rates have been increasing in recent years, especially in developing countries, and the reasons for this deserve further investigation. In this study, we examine the relationship between short videos and body weight using large-scale micro-survey data from China. Our findings show that short videos increase BMI and the probability of obesity. The negative health impact is substantial, with an estimated cost of CNY 7.53 billion in additional healthcare expenses nationwide. It can explain about 27 % of the growth in China's obese population from 2020 to 2022, suggesting that it has become a major new driver of obesity.

We identify the primary mechanisms driving this effect: short videos are addictive, leading to increased time spent on mobile devices especially for entertainment, and a reduction in time spent exercising. Furthermore, we find no evidence to support the idea that short videos promote exercising.

Moreover, there are no other mechanisms, including mental health, sleep quality, and health-related habits. Perhaps it is because they are offset by both positive and negative influences. Our heterogeneity analysis reveals that the negative effect is pronounced for urban, male, and individuals with lower education levels or aged over 35.

Given the substantial medical costs associated with short video-induced obesity, we recommend that regulatory policies be introduced to address algorithmic addiction on short video platforms. Specifically, platform algorithms should be adjusted to reduce excessive user engagement and promote healthier usage patterns. In parallel, public health campaigns should emphasize the importance of physical activity and raise awareness about the health risks of watching short videos for a long time. These combined efforts could help mitigate the negative externalities of short videos on public health.

While our study contributes to the understanding of the impact of short videos on body weight, it has some limitations. First, due to data constraints, we were unable to explore other potential mechanisms, such as prolonged sitting. Second, the data covers only a two-year period, making it difficult to assess the long-term effects. It is possible that the beneficial effects of short videos on exercise may manifest over a longer timeframe. Future research could benefit from more detailed, long-term data to further explore the mechanisms through which short videos contribute to weight gain.

CRedit authorship contribution statement

Qi Wang: Writing – review & editing, Visualization, Formal analysis, Conceptualization. **Chenxuan Chen:** Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

⁴ According to China's census data, the population aged 16 and above was 1159.35 million in 2020 and 1155.6 million in 2022 respectively. Therefore, the increase in the number of obese people from 2020 to 2022 was 11.28 million ($= 1155.6 \text{ million} \times 8.3\% - 1,159.35 \text{ million} \times 7.3\%$).

Acknowledgements

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Appendix

Table A1
Definitions of variables

Variable	Definition
BMI	A variable calculated according to height and weight: $BMI = weight / height^2$.
Obesity	A dummy variable that equals one if the respondent's BMI is more than 28 and zero otherwise.
Short videos	A dummy variable that equals one if the respondent watches short videos and zero otherwise.
Age	A variable that indicates how old the respondent is.
Age ²	A variable indicating age squared of the respondent.
Gender	A dummy variable that equals one if the respondent is male and zero otherwise.
Education	A categorical variable representing education level, 0 is no schooling or illiteracy, 1 is nursery school education, 2 is kindergarten education, 3 is primary school education, 4 is junior high school education, 5 is senior high school/technical school and vocational high school education, 6 is junior college education, 7 is university education, 8 is master's degree, and 9 is doctor's degree.
Urban	A dummy variable that equals one if the respondent lives in city and zero otherwise.
Medical insurance	A dummy variable that equals one if the respondent has medical insurance and zero otherwise.
ln(family income)	The natural logarithm of total household income.
Family size	The number of family members in the household.

Table A2
Controlling for total mobile device use

	(1)	(2)
VARIABLES	BMI	Obesity
Short videos	0.115** (0.055)	0.022*** (0.007)
Controls	Yes	Yes
Individual FE	Yes	Yes
Year FE	Yes	Yes
Observations	16,388	16,388
Adjusted R-squared	0.866	0.667

This table reports the baseline results after controlling for total mobile device use. The dependent variables are BMI in column (1), and an obesity indicator in column (2). All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Table A3
Two-Stage Selection Issue

	(1)	(2)	(3)	(4)
VARIABLES	BMI	BMI	Obesity	Obesity
Short videos		0.117** (0.054)		0.022*** (0.007)
Mobile	0.001 (0.443)	-0.009 (0.443)	0.013 (0.016)	0.011 (0.016)
Controls	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	16,388	16,388	16,388	16,388
Adjusted R-squared	0.865	0.866	0.667	0.667

This table reports the baseline results after solving two-stage selection issue. The dependent variables are BMI in columns (1) and (2), and an obesity indicator in columns (3) and (4). All the results control the individual and year fixed effects. Standard errors are cluster at the province-level in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Data Availability

The authors do not have permission to share data.

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